LETAO (JENNA) CHEN

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EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA Los Angeles, CA
Ph.D. in Computer Science (Sep 2024 - present)

NEW YORK UNIVERSITY, The Courant Institute of Mathematical Sciences New York, NY
M.S. in Scientific Computing (May 2024)

UNIVERSITY OF TORONTO Toronto, ON
Honours Bachelor of Science
Mathematics & Its Application in Stats & Probability Specialist, Computer Science Minor (June 2022)

RESEARCH INTERESTS

Geometry processing, surface reconstruction, computer graphics, scientific computing, numerical integral equation

RESEARCH EXPERIENCE

UNIVERSITY OF SOUTHERN CALIFORNIA

Signed Distance Function (SDF) Compression
Ph.D. Research, supervised by Prof. Oded Stein and Yue Wang (Feb 2026 - present)

- Applied additively weighted Voronoi diagrams (Apollonius diagrams) to accelerate SDF validity checks
- Used Apollonius diagrams to identify and prune redundant SDF samples based on the size of exposed arcs in their spherical representation, balancing data reduction with surface reconstruction quality

NEW YORK UNIVERSITY

Boundary Integral Methods for 2D Boundary Value Problems on Complex Domains
Research Assistant, supervised by Prof. Denis Zorin (Nov 2022 - present)

- Implemented and evaluated BIE solvers for Laplace's equation on smoothed boundaries, with quantitative comparison against finite element method (FEM) solutions.
- Developed robust preprocessing pipelines for complex boundaries, including curvature-based smoothing, corner rounding and merging, intersection detection, and adaptive resampling.

STANFORD UNIVERSITY

Generation of Synthetic Datasets for Cardiovascular Models
Research Assistant, supervised by Prof. Eric Darve and Abeynaya Gnanasekaran (Oct 2021 - Aug 2022)

- Applied mesh deformation techniques using Laplacian-based formulations to synthesize anatomically plausible model variations.
- Generated high-quality volumetric meshes from surface representations for cardiovascular geometry modeling.

WORKING PAPER

Status: Conditionally **accepted** at *ACM Transactions on Graphics (TOG)*

TEACHING EXPERIENCE

UNIVERSITY OF SOUTHERN CALIFORNIA

- CSCI 585: Database Systems (Spring 2025, Fall 2026)

COMPUTATIONAL SKILLS

Programming Languages: C++, Python, MATLAB, C
Technical Tools: CUDA, OpenMP, PyTorch